

Certified by the Ceiling: Saturated Benchmarks Supply the No-Loss Claims Behind VLA Acceleration

Nihal Gunukula
Purdue University
ngunukul@purdue.edu

Abstract—Papers that make vision-language-action (VLA) policies cheaper to run justify their speedups by showing no loss on LIBERO, where base policies typically succeed in more than 90% of episodes. We show that such no-loss verdicts are largely supplied by tasks both settings already solve. With binary outcomes two settings can disagree on at most the episodes where one of them fails, so saturated tasks force agreement and tighten an absolute-difference interval without informing it about the tasks where policies fail. On released $\pi_{0.5}$, one denoising step is equivalent to ten within ± 5 points on an eight-task battery but inconclusive on the tasks with headroom; of 18 certificates among 79 paired comparisons, saturated tasks keep 17 and informative tasks 1 at equal episode counts. Where failures are common, one step costs π_0 7.9 points and $\pi_{0.5}$ 7.1 points on the hardest camera-viewpoint shifts. Among 86 VLA efficiency papers claiming parity on LIBERO, the median base policy succeeds in 96.4% of episodes, where a ± 5 -point margin tolerates $2.35\times$ the base’s failures, and 15 report any uncertainty, per-task result, or paired analysis. We give a three-part report for no-loss claims and a tool that computes it.

I. INTRODUCTION

Making VLA policies cheaper at inference is now a research area of its own: fewer denoising steps [1], [2], caching [3], adaptive computation [4], token pruning, and quantization, applied to released policies such as π_0 , $\pi_{0.5}$, and SmolVLA [5], [6], [7]. Their shared evidence is parity with the base policy on LIBERO [8]; released $\pi_{0.5}$, for example, loses 0.2 points at a single denoising step [3].¹

Claim. Such no-loss verdicts are largely supplied by tasks both settings already solve, and so say little about the tasks on which policies fail. The underlying leniency of absolute margins at low event rates is known in non-inferiority trials [9], [10], and in language-model compression aggregate parity is known to hide per-example changes [11], [12], [13], with a recent audit of equivalence claims for compressed LLMs [14]. We isolate what those works do not: on a saturated benchmark, disagreement is bounded by failures, so the certificate is decided by the few tasks that still fail, and it does not transfer to them. **Contributions:** (1) a quantification of this dilution for VLA evaluation, with an equal-size control over task subsets (Sec. III); (2) evidence of where the shortcut costs something once failures are common (Sec. IV); (3) an audit of 86 VLA

efficiency papers showing their claims sit in this regime, and a three-part report with a tool that computes it (Secs. V, VI).

II. SETUP

Released $\pi_{0.5}$ and π_0 LIBERO checkpoints (openpi) and SmolVLA, served through one harness; 31,858 per-episode records are released. The battery is eight LIBERO-10 tasks $\times$ 20 initial states $\times$ 3 sampling seeds ($n=480$ per setting). A design written before any grid episode classified four tasks as informative (development-scan success in $[0.5, 0.95]$: tasks 0, 2, 8, 9) and four as saturated, for a different study; using it here is our choice after the fact. Under ten steps the informative tasks succeed in 95, 97, 80, and 95% of episodes and the saturated ones in 97–100%, so the split rests chiefly on task 8. Shifts are fixed LIBERO-Plus sets [15]: 312 object-layout variants and 419 camera-viewpoint variants (two draws each), each with benchmark-defined difficulty levels 1–5. “One step” is a single Euler step of the released flow head, against ten, both at execution horizon 10. Comparisons are paired on (initial state, seed) or (variant, draw) and tested with exact McNemar [16]; intervals are Newcombe’s paired score intervals [17]. A comparison is a significant difference if $p < 0.05$, otherwise equivalent if its 95% interval lies within ± 5 points (stricter than two one-sided tests [18]), otherwise inconclusive.

III. SATURATED TASKS SUPPLY THE CERTIFICATE

One contrast. On the battery, $\pi_{0.5}$ succeeds in 0.965 of episodes at one step and 0.946 at ten (23–14 discordant pairs, $p=0.19$): equivalent, $[-0.7, +4.5]$. Admitting tasks from the hardest to the easiest, the interval fits inside the margin only at the end (Fig. 1a). On the four informative tasks it is $[-2.0, +7.1]$, inconclusive; nothing here shows a cost, but nothing certifies its absence either.

Every contrast. The released grids hold 79 paired comparisons between dial settings on identical episodes ($\pi_{0.5}$ at two environment seeds, and π_0). On the battery 18 certify equivalence (all on $\pi_{0.5}$, over 13 distinct settings, so not independent) and 59 differ (48 after Holm). At the same 240 pairs (Fig. 1b), the four saturated tasks keep 17 of the 18 certificates but only 38 of the 59 differences; the four informative tasks keep 1 and 57. The informative half is the unique minimum of all 70 subsets, and subsets without task

¹Per-episode logs, audit extractions, and code: <https://github.com/nihalgunu/roborigor>

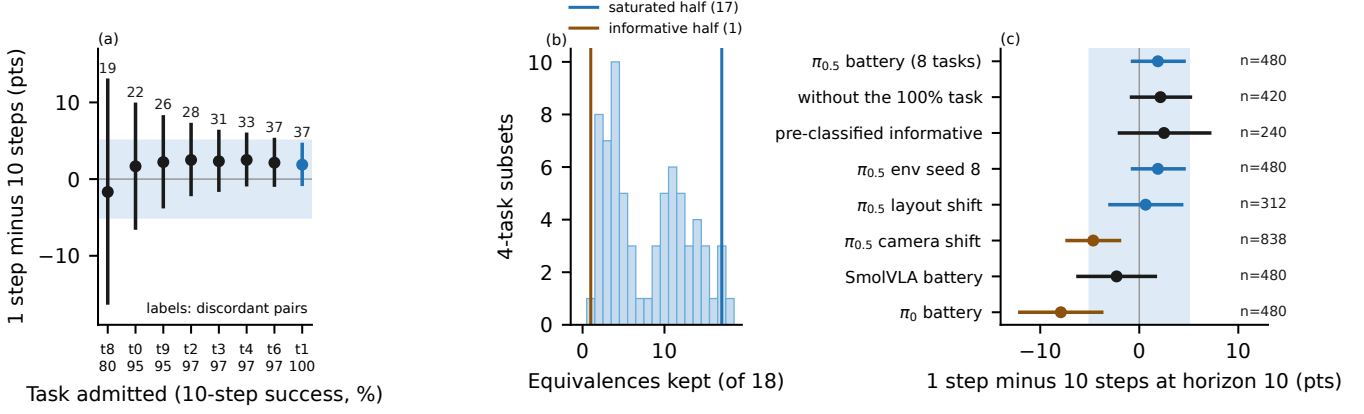


Fig. 1. Where equivalence certificates come from: one denoising step minus ten at execution horizon 10 (paired Newcombe 95% intervals; shaded: ± 5 -point margin). (a) $\pi_{0.5}$ on the LIBERO-10 battery, admitting tasks from least to most successful under ten steps (ticks: task, ten-step success; labels: discordant pairs). The interval enters the margin only once the near-saturated tasks are in. (b) The 18 comparisons in our grids that certify equivalence on the battery, re-tested on each of the 70 four-task subsets (240 pairs each). (c) The contrast beyond the saturated battery (SmolVLA against its served default); brown: significant difference, blue: equivalent, black: inconclusive.

8 keep a median of 12 certificates against 4 with it. The asymmetry holds under unpaired Newcombe (14 vs. 0), paired Wald (17 vs. 3), a 90% rule (18 vs. 5), and a ± 7 margin (19 vs. 10).

Why. For binary outcomes the probability that two settings disagree on an episode is at most the sum of their failure rates, $f_A + f_B$; on the battery that bound is 9.0% and the observed disagreement 7.7%. Saturated tasks therefore contribute pairs that cannot disagree: they leave the discordant counts b, c unchanged while the difference $(b - c)/n$ and its standard error shrink as $1/n$, faster than the $1/\sqrt{n}$ of informative pairs. Equivalently, the pooled difference is the task-weighted mean of per-task differences: if a share w of pairs comes from tasks that can fail, a pooled certificate at $\pm m$ is compatible with an average loss of up to m/w on those tasks. Task 8 carries 46% of the battery’s ten-step failures but one eighth of its pairs, so a ± 5 battery certificate is compatible with a 40-point loss on it. Certifying ± 5 on the informative tasks directly would take about 520 paired episodes against the 240 the battery allots.

IV. BEYOND THE SATURATED BATTERY

On released π_0 , whose battery rate is lower, one step costs 7.9 points (0.773 vs. 0.852, 33–71, $p=0.00025$, $[-12.1, -3.8]$): whether one step suffices depends on the policy. SmolVLA is inconclusive ($[-6.2, +1.6]$). Under object-layout shift, where $\pi_{0.5}$ still succeeds in 91% of episodes, one step is equivalent ($[-3.0, +4.3]$). Under camera-viewpoint shift it costs 4.7 points overall (45–84, $p=0.0008$, $[-7.3, -2.0]$), and the cost sits where the benchmark places its hardest shifts: at levels 4–5 one step loses 7.1 points (31–74, $[-10.4, -3.8]$), while at levels 1–3 the contrast is inconclusive ($[-2.5, +6.0]$). ActionCache likewise reports $\pi_{0.5}$ collapsing at one step on the harder VLABench [3].

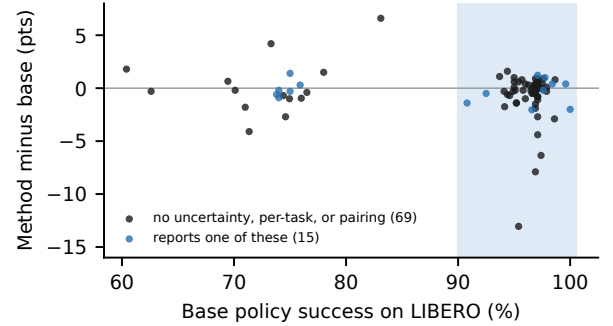


Fig. 2. No-loss claims in the 84 audited VLA efficiency papers with numeric LIBERO rates. Shaded: base at or above 90%. Blue: reports uncertainty, per-task results, or a paired analysis.

V. THE LITERATURE’S NO-LOSS CLAIMS

Under a written search and extraction protocol (354 candidates, two-stage screening), 86 VLA inference-efficiency papers claim parity or acceptable loss against their own base policy on LIBERO, spanning step reduction, distillation, caching, token pruning, quantization, early exit, and speculative decoding (Fig. 2). The base succeeds in at least 90% of episodes in 64 of the 84 claims with numeric rates (median 96.4%). At that median a ± 5 -point margin tolerates $2.35\times$ the base’s failures, and in 51 of 83 claims it tolerates at least double. Only 15 of 86 report any uncertainty, per-task result, or paired analysis, and 45% state no usable episode count. The same comparisons do not lose more on the harder benchmarks the papers also report: the loss is larger there in 9 of 30 papers and smaller in 21 (sign test $p=0.04$), though those benchmarks are author-selected and often small real-robot tests where lower latency itself helps. The audit thus shows that the field’s certificates are uninformative, not that the certified shortcuts are costly. Extraction was LLM-assisted; text search

of a seeded 20% sample confirmed 51 of 54 fields, and manual checks the rest.

VI. A REPORT FOR NO-LOSS CLAIMS

Relative margins remove the dilution but fail the other way: the failure-rate ratio ignores solved tasks, yet under camera shift, where most episodes fail, it is 1.08 [1.03, 1.14] and would certify non-inferiority at a 1.5 bound although failures rise significantly. We therefore recommend reporting (i) the discordant pairs with the exact conditional odds-ratio interval; (ii) the absolute interval on tasks declared informative before the experiment; and (iii) a failure-rate-ratio interval against a declared bound, certifying only if (ii) lies inside the margin and (iii) lies below the bound. `roborigor` report computes all three from paired records and warns when a pooled certificate does not hold on the informative tasks.

Limitations. Simulation only, on LIBERO-family benchmarks and three flow-matching policies. The certificates in our grids come from one policy and share arms, the informative/saturated split rests chiefly on one task and was made for a different study, and we re-test one acceleration; the audit shows other families make the same kind of claim but does not re-test them.

Conclusion. A no-loss verdict on a saturated benchmark is valid for the average it certifies and uninformative about the tasks on which policies fail. The VLA efficiency literature makes its claims in exactly this regime; they should come with discordant pairs, results on informative tasks, and a relative margin.

REFERENCES

- [1] W. Luan, J. Li, W. Zhao, W. Zhang, T. Wu, and R. Ma, “SnapFlow: One-step action generation for flow-matching VLAs via progressive self-distillation,” *arXiv:2604.05656*, 2026.
- [2] Y. Chen *et al.*, “Let it be simple: One-step action generation for vision-language-action models,” *arXiv:2606.05737*, 2026.
- [3] R. Oi, H. Otsuka, K. Matsushima *et al.*, “ActionCache: Training-free acceleration for vision-language-action models with action caching and refinement,” *arXiv:2607.06370v2*, 2026.
- [4] S. Han, Y. Han, and Y. Yi, “AdaVLA: Adaptive step flow matching for training-free acceleration of vision-language-action models,” *arXiv:2608.29208*, 2026.
- [5] K. Black *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv:2410.24164*, 2024.
- [6] Physical Intelligence, “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” *arXiv:2504.16054*, 2025.
- [7] M. Shukor *et al.*, “SmolVLA: A vision-language-action model for affordable and efficient robotics,” *arXiv:2506.01844*, 2025.
- [8] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*, 2023.
- [9] M. Quartagno, A. S. Walker, A. G. Babiker, R. M. Turner, M. K. B. Parmar, A. Copas, and I. R. White, “Handling an uncertain control group event risk in non-inferiority trials: Non-inferiority frontiers and the power-stabilising transformation,” *Trials*, vol. 21, p. 145, 2020.
- [10] B. Redfors, “A case for using relative rather than absolute noninferiority margins in clinical trials,” *JACC: Advances*, vol. 3, no. 4, p. 100913, 2024.
- [11] A. Dutta, S. Krishnan, N. Kwatra, and R. Ramjee, “Accuracy is not all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [12] S. Hooker, A. Courville, G. Clark, Y. Dauphin, and A. Frome, “What do compressed deep neural networks forget?” *arXiv:1911.05248*, 2019.
- [13] J. Kübler, K. Budhathoki, M. Kleindessner *et al.*, “When LLMs get significantly worse: A statistical approach to detect model degradations,” *arXiv:2602.10144*, 2026.
- [14] A. Singh, “Certifying compressed language models: An audit and a statistical toolkit,” *arXiv:2608.15046*, 2026.
- [15] S. Fei *et al.*, “LIBERO-Plus: In-depth robustness analysis of vision-language-action models,” *arXiv:2510.13626*, 2025.
- [16] Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” *Psychometrika*, vol. 12, pp. 153–157, 1947.
- [17] R. G. Newcombe, “Improved confidence intervals for the difference between binomial proportions based on paired data,” *Statistics in Medicine*, vol. 17, no. 22, pp. 2635–2650, 1998.
- [18] D. J. Schuurmann, “A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability,” *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987.